Fall 2025 CPSC 332-11 21307 – File Structures and Databases

Project Instructions

Instructor: Hokseng Hun

[Version 2025.08.26]

Important Date

1. Part A – Due on Monday, Nov 11, 2025, at 11:59 pm
2. Part B – Due on Monday, Nov 23, 2025, at 11:59 pm
3. Presentation - Due on Monday, Dec 8, 2025, at 5:00 pm
4. Part C – Due on Monday, Dec 14, 2025, at 1:59 pm

Project submissions should be made through Canvas. Students are required to work in groups to complete the project, submitting a final report, a demo, and a presentation. Additionally, each team must submit a peer review, where they will assign a percentage rating of each group member's contribution to the project. Ideally, all members should contribute equally. However, if a student is assessed to have participated less, their grade will be adjusted accordingly. For example, if a student only contributes 50% of their required tasks and receives a 50% review from their peers, they will only receive 50% of the total possible score.

Note

Submissions made via email will not be accepted. Only those submitted through Canvas will be graded.

Total Score

The project consists of three main parts, with a total of 125 points, and offers extra credit opportunities for additional functionality.

Database Application Project

As a freelance software development team, you will receive a project request from a client to develop a movie ticket reservation software system.

The client will provide a set of business requirements that their company expects you to use to develop the desired application. Meeting these requirements is a top priority, as the client will be strict in ensuring they are fulfilled. The project you are commissioned for will involve the following:

Building an ER Diagram based on the provided requirements.

- Building a Relational Model based on your ER Diagram.
- Normalize your database to 3NF.
- Building the Physical Model inside your environment.
- Populate your database with sample data.
- Create a demo application to showcase your database design in PHP and MySQL

Client Business Requirements

Your client is ABC Entertainment Company, a popular cinema chain across the country. They've asked your team to build a cinema management system with both a database and a web app. The database should keep track of everything— cinemas, auditoriums, seats, movies, and screening times.

Business Rules

1. The company operates numerous cinemas across the country.
2. Each cinema record includes its name, address, and employees and roles (theater manager, technician, ticketing, concessions, ushers)
3. Employees should be able to search movies and showtimes by location, date and time.
4. System access requires employees log in using email and password.
5. Employee profiles must include:
    - First and last name
    - Address
    - Phone number
    - Email
    - Password
6. Address must include: street number, City, State, and Zip code.
7. Passwords must be at least eight characters long and include at least one uppercase letter, one lowercase letter, one number, and one special character.
8. Password encryption is optional.
9. Auditorium details must include:
    - Cinema
    - Name
    - Capacity
    - 3D support
    - IMAX support
10. Auditorium seats are numbered and arranged in rows.
11. The movie records must include:
    - Title
    - Description

- year
- duration
- Rating (G, PG, PG-13, R)
- Release date
- Genres (Action, animation, comedy, family, horror, thriller)

12. Showtime must have:
    - Auditorium
    - Movie
    - Start and end time
    - Format (regular or 3D)
13. All records must have a creation timestamp.

Additional Submission Requirements for the Project

1. Write a SQL script to create a database in MySQL Server, ensuring that the script successfully creates the database and all associated tables.
2. Insert sample data into the tables.
3. Create a VIEW showing cinema name, auditorium name, and movie title for movies with a duration longer than 120 minutes, and group the results by genre.
4. Create a VIEW displaying cinema name, address, auditorium name, capacity and 3D-support screenings.
5. Create a VIEW listing all movies shown more than 3 times per day.

Extra credit

Enhance your website with additional functionalities and ensure it has a professional appearance. You may utilize free or open-source CSS templates available online to achieve this.

Peer Evaluation

List the names of your team members along with their contributions.

- John, 100%
- Jason, 67%.

If you earn 35 out of 40 points on Project Part A, John will receive 35 points, whereas Jason will receive only 23.5 points.

Submission 1 – Project Part A [40 points]

Submit your Entity Relationship Conceptual Schema Diagram and Relational Database Schema. Make sure the following are shown:

- Tables
- Primary keys
- Foreign keys
- Relationships
- Any Constraints

Submission 2 – Project Part B [30 points]

Submit your development files, presentation, and final report (one per group.) For your development files, include an SQL Script of the populated database you created in your environment.

Submission 3 – Project Part C [30 points]

Your final report will outline the process of taking the requirements and showing the design process of the database. The recommended outline is as follows:

> **Title (Application Name, Project Team Name, Class, Date, Team Members)**
>
> **Introduction**
>
> **Database Design Process**
>
> > Requirements
> >
> > ER Diagram
> >
> > Relational Model
> >
> > Physical Model
>
> **Application Design**
>
> > Overview
> >
> > SQL Queries
> >
> > Final Product
>
> **Summary**
>
> **Peer Evaluation**

Presentations [25 points]

You'll need to present your demo in person. Please submit them ahead of time so we can make sure your demo runs smoothly and your presentation shows off all the key features of your app.

Extra credit [5 points]

The requirements include opportunities for extra credit. Review them carefully to determine if you would like to incorporate these features into your product.